



Jenkins Setup

Introduction

Jenkins is a **powerful automation server** used for **Continuous Integration (CI)** and **Continuous Deployment (CD)**.

- How do you install and configure Jenkins?
- What are the system requirements for Jenkins?
- How do you access Jenkins for the first time?

This guide provides a **step-by-step approach to setting up Jenkins**.

Section 1: Learn

System Requirements for Jenkins

Component	Minimum Requirement
RAM	1 GB (2 GB recommended)
Disk Space	50 GB (recommended for large projects)
Operating System	Windows, macOS, Linux (Ubuntu, CentOS, RedHat)
Java Version	Java 11 or later
Network Port	Default: 8080 (can be changed)

Why Set Up Jenkins?

- **Automates software builds and tests** to improve development speed.
- **Provides a centralized platform** for managing CI/CD pipelines.
- **Supports multiple integrations** with DevOps tools like Git, Docker, Kubernetes, and AWS.



Interesting Fact

Jenkins was **originally developed in 2004** as **Hudson** and later renamed **Jenkins in 2011** after Oracle acquired Sun Microsystems.

Section 2: Practice

1. Installing Jenkins on Ubuntu

Step 1: Update System Packages

```
sudo apt update && sudo apt upgrade -y
```

Step 2: Install Java (Required for Jenkins)

```
sudo apt install openjdk-11-jdk -y
```

Verify Java installation:

```
java -version
```

Step 3: Add Jenkins Repository and Install Jenkins

```
wget -q -O - https://pkg.jenkins.io/debian/jenkins.io.key | sudo apt-key add -  
sudo sh -c 'echo deb http://pkg.jenkins.io/debian-stable binary/ >  
/etc/apt/sources.list.d/jenkins.list'  
sudo apt update  
sudo apt install jenkins -y
```



Step 4: Start and Enable Jenkins

```
sudo systemctl start jenkins  
sudo systemctl enable jenkins
```

Check Jenkins status:

```
sudo systemctl status jenkins
```

2. Accessing Jenkins for the First Time

Step 1: Open Jenkins in a Web Browser

1. Open a browser and go to:

```
http://localhost:8080
```

2. If running on a remote server, replace **localhost** with the server's IP address.

Step 2: Retrieve the Initial Admin Password

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

Copy and paste the password into the Jenkins setup page.

Step 3: Install Suggested Plugins

1. Click **"Install Suggested Plugins"** to install common plugins automatically.
2. Wait for the installation to complete.



Step 4: Create an Admin User

1. Enter a **Username, Password, Full Name, and Email Address**.
2. Click **Save and Continue**.

Jenkins is now **ready for use!**

3. Configuring Jenkins Settings

Step 1: Open Global Configuration

1. Click **Manage Jenkins → Configure System**.
2. Set **Jenkins URL** to:

```
http://your-server-ip:8080
```

Step 2: Configure Security

1. Click **Manage Jenkins → Configure Global Security**.
2. Enable **"Jenkins Own User Database"**.
3. Click **"Enable security"** and set permissions.

Step 3: Set Up Email Notifications

1. Click **Manage Jenkins → Configure System → E-mail Notification**.
2. Enter SMTP settings to enable email alerts.

Click **Save**.

Why is this important?

- Ensures Jenkins is secure and configured correctly before creating jobs.
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Section 3: Know More

Frequently Asked Questions

1. What is the default Jenkins port?
 - Jenkins runs on port 8080 by default.
2. How do I change the Jenkins port?
 - Edit the Jenkins configuration file:

```
sudo nano /etc/default/jenkins
```

- Change `HTTP_PORT=8080` to another port, e.g., `9090`, then restart Jenkins:

```
sudo systemctl restart jenkins
```

3. What if Jenkins doesn't start after installation?
 - Check logs using:

```
sudo journalctl -u jenkins -f
```

4. Where is Jenkins data stored?
 - Jenkins stores data in `/var/lib/jenkins/`.
5. How can I back up Jenkins?
 - Copy the `/var/lib/jenkins` directory to a backup location.

Conclusion

Jenkins setup is the first step towards automating CI/CD workflows.

Start by installing Jenkins, configuring settings, and securing access, and soon you'll be ready to build automation pipelines!